Temperature and Power in a Synthetic Sensor Dataset: A Typesetting Demonstration

Muhammad Uzair Javed

***Abstract*—This paper demonstrates an IEEE-style workflow using 2,880 synthetic hourly observations from three example sites. A hinge regression is fitted by least squares, with its threshold selected by a grid search. The in-sample coefficient of determination is $R^2 = 0.850$; it describes the fit to the generated rows and is not a measure of performance on new data. A separate block diagram and second-order response plot illustrate feedback-control notation. No physical system or controller is evaluated.**

***Index Terms*—LaTeX typesetting, synthetic data, hinge regression, TikZ, reproducible figures.**

I. Introduction

This example uses a small synthetic dataset to demonstrate an academic paper workflow. The data script creates the rows; the figure script prepares charts and a LaTeX table. The document uses pandas [1], Matplotlib [2], and seaborn [3]. The colour palette follows guidance for accessible scientific graphics [4].

Section II describes the regression specification. Section III contains separate control-theory illustrations, and Section IV summarizes the generated data.

II. Demand Model

Let T_k denote temperature and P_k power at hour k. The example uses the hinge model in Equation (1):

$$P_k = \beta_0 + \beta_1 \max(T_k - \tau, 0) + \varepsilon_k, \qquad \varepsilon_k \sim \mathcal{N}(0, \sigma^2). \quad (1)$$

For each candidate threshold τ, the intercept and slope are found by ordinary least squares. The threshold is selected from a fixed grid by minimizing the residual sum of squares.

Stacking the observations gives $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. The least-squares estimator and its covariance are given in Equation (2), with the two component equations in Equations (2a) and (2b).

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}, \quad (2a)$$

$$\operatorname{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1}. \quad (2b)$$

For the two-column design with $h_k = \max(T_k - \tau, 0)$, the normal equations are shown in Equation (3):

$$\begin{bmatrix} N & \sum h_k \\ \sum h_k & \sum h_k^2 \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} = \begin{bmatrix} \sum P_k \\ \sum h_k P_k \end{bmatrix}. \quad (3)$$

This paper is a portfolio typesetting example. The observations are synthetic and do not represent a real monitoring site. Source: https://github.com/uzairjaved-prog/Academic-typesetting-portfolio.

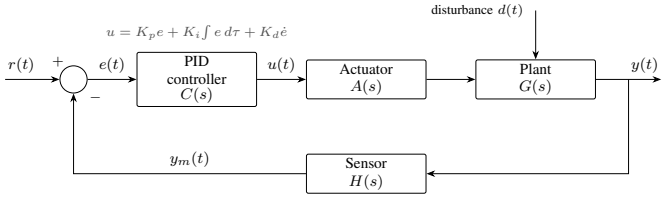


Fig. 1: Block diagram of a feedback loop with PID control.

The residual variance and the variance of the slope estimator are summarized in Equation (4):

$$\hat{\sigma}^2 = \frac{1}{N-2} \sum_{k=1}^{N} (P_k - \hat{\beta}_0 - \hat{\beta}_1 h_k)^2,$$
$$\operatorname{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_k (h_k - \bar{h})^2}. \quad (4)$$

Under the model assumptions, the estimator is unbiased; the resulting identities are listed in Equation (5):

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}, \qquad \operatorname{tr}\Big[\operatorname{Cov}(\hat{\boldsymbol{\beta}})\Big] = \sigma^2 \operatorname{tr}\big[(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\big]. \quad (5)$$

A standard treatment of estimation bounds is given by Kay [5].

III. Control-Theory Illustrations

The following diagrams are separate illustrations. They are not fitted to the synthetic observations, and no controller is tested against a physical plant.

The block diagram in Figure 1 shows a feedback loop with a PID controller. Its control signal is defined in Equation (6):

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \dot{e}(t). \quad (6)$$

The second-order plant model used for the response illustration is

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \quad (7)$$

Figure 2 compares its unit-step response at three damping ratios. Lower damping produces greater overshoot in this plot. A small supervisory finite-state diagram appears in Figure 3; the state-machine notation is described by Hopcroft et al. [6]. General feedback-control background is available in Ogata [7].

IV. Results

The generated dataset contains 2,880 hourly rows from three example sites. Temperature and power have Pearson correlation $r = 0.86$. The hinge regression has in-sample $R^2 = 0.850$. Both values describe this generated dataset; neither measures performance on independent observations.

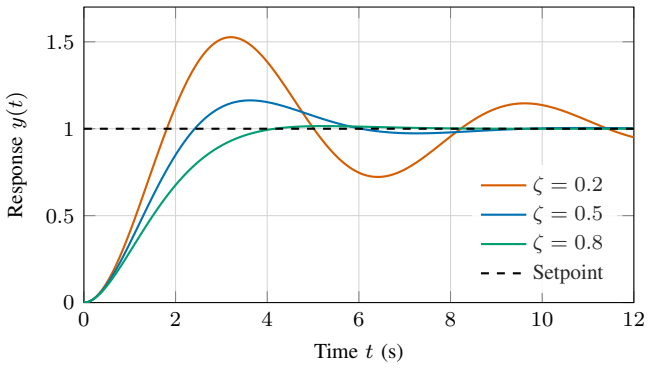


Fig. 2: Unit-step response of the second-order model in Equation (7) for three damping ratios.

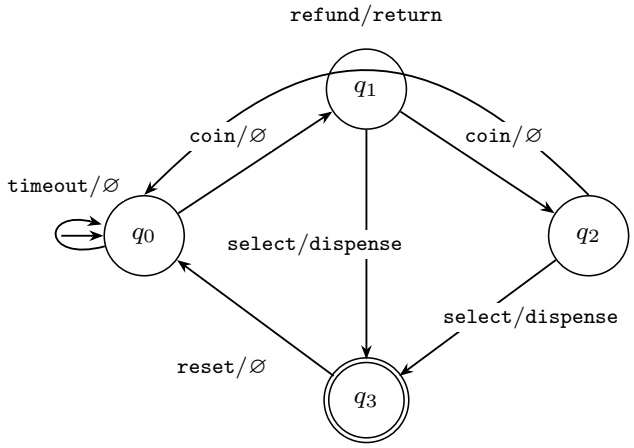


Fig. 3: Example supervisory finite-state machine.

TABLE I: Summary statistics by example site.

Site	n	$\bar{T}$ (°C)	$\bar{P}$ (W)	s_P (W)	$\bar{a}$ (g)
Site A	1151	19.87	133.74	16.60	0.377
Site B	989	20.18	139.18	16.92	0.385
Site C	740	19.87	131.00	16.21	0.383

The four-panel summary is shown in Figure 4. Panel Figure 4a shows daily mean power, Figure 4b shows pairwise correlations, and Figure 4c compares the generated site distributions. Panel Figure 4d shows the fitted hinge curve and the dashed linear baseline. The per-site counts, means, and standard deviations appear in Table I.

The hinge coefficients are solved in closed form by least squares for each candidate threshold. The grid search selects the threshold with the smallest residual sum of squares; the coefficient fit itself does not use an iterative convergence procedure.

V. Conclusion

This paper demonstrates a reproducible route from generated CSV data to an IEEE-style PDF. The regression fit, correlation, figures, and table all come from the same synthetic dataset. The controller and step-response diagrams are separate mathematical illustrations. They do not show a controller tested on the data or on a real system.

References

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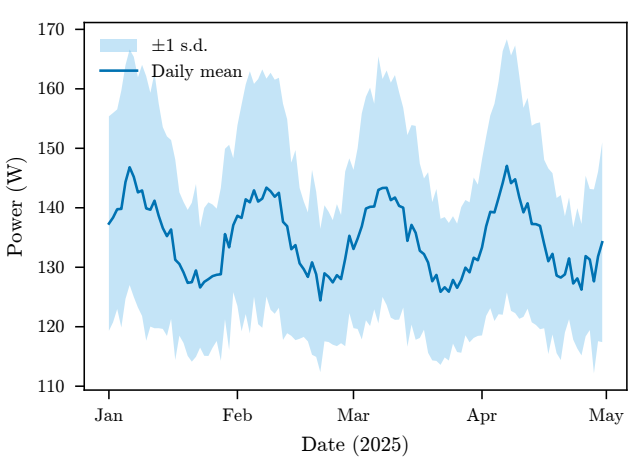
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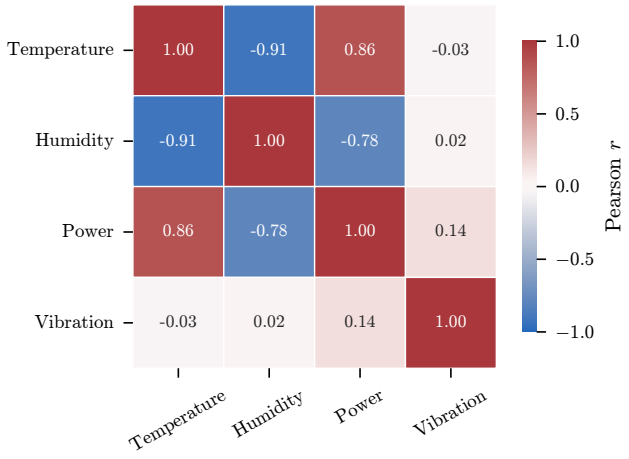
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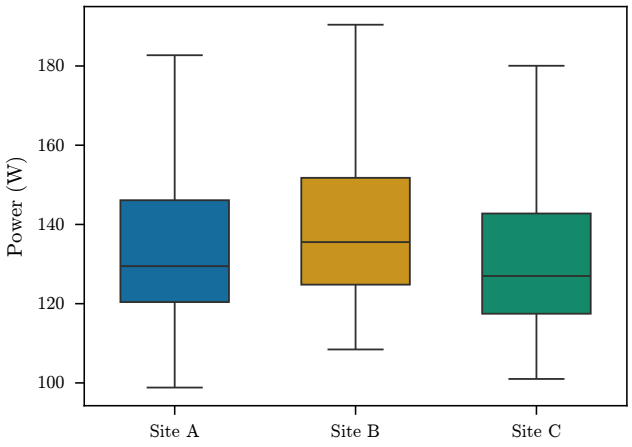
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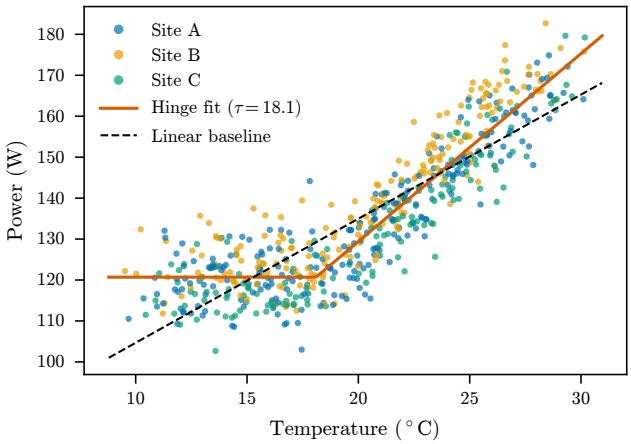
(a) Daily mean power with a one-standard-deviation band.



(b) Correlation matrix.



(c) Power distribution by site.



(d) Power against temperature with hinge fit and linear baseline.

Fig. 4: Exploratory summaries from the synthetic sensor dataset.